

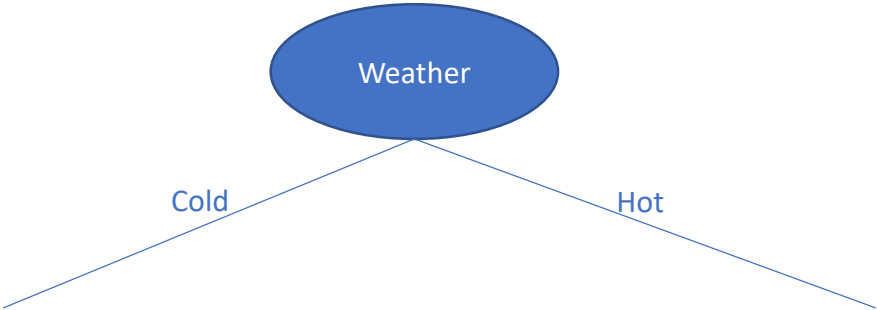
Naloga 1 (demonstracija)

Zgradite odločitveno drevo za napovedovanje ciljne spremenljivke **Order**. Za izbiro najboljšega atributa uporabite **informacijski prispevek**.

TimeOfDay	Weather	Order
Evening	Cold	Coffee
Morning	Cold	Coffee
Evening	Cold	Coffee
Morning	Hot	Coffee
Evening	Hot	Beer
Evening	Cold	Beer
Morning	Hot	Beer
Evening	Cold	Coffee
Morning	Cold	Coffee
Morning	Cold	Coffee
Evening	Cold	Beer
Morning	Cold	Coffee
Morning	Cold	Coffee
Evening	Hot	Beer
Morning	Cold	Coffee
Morning	Hot	Beer

$\text{InfGain}(\text{TimeOfDay}) = 0.093$

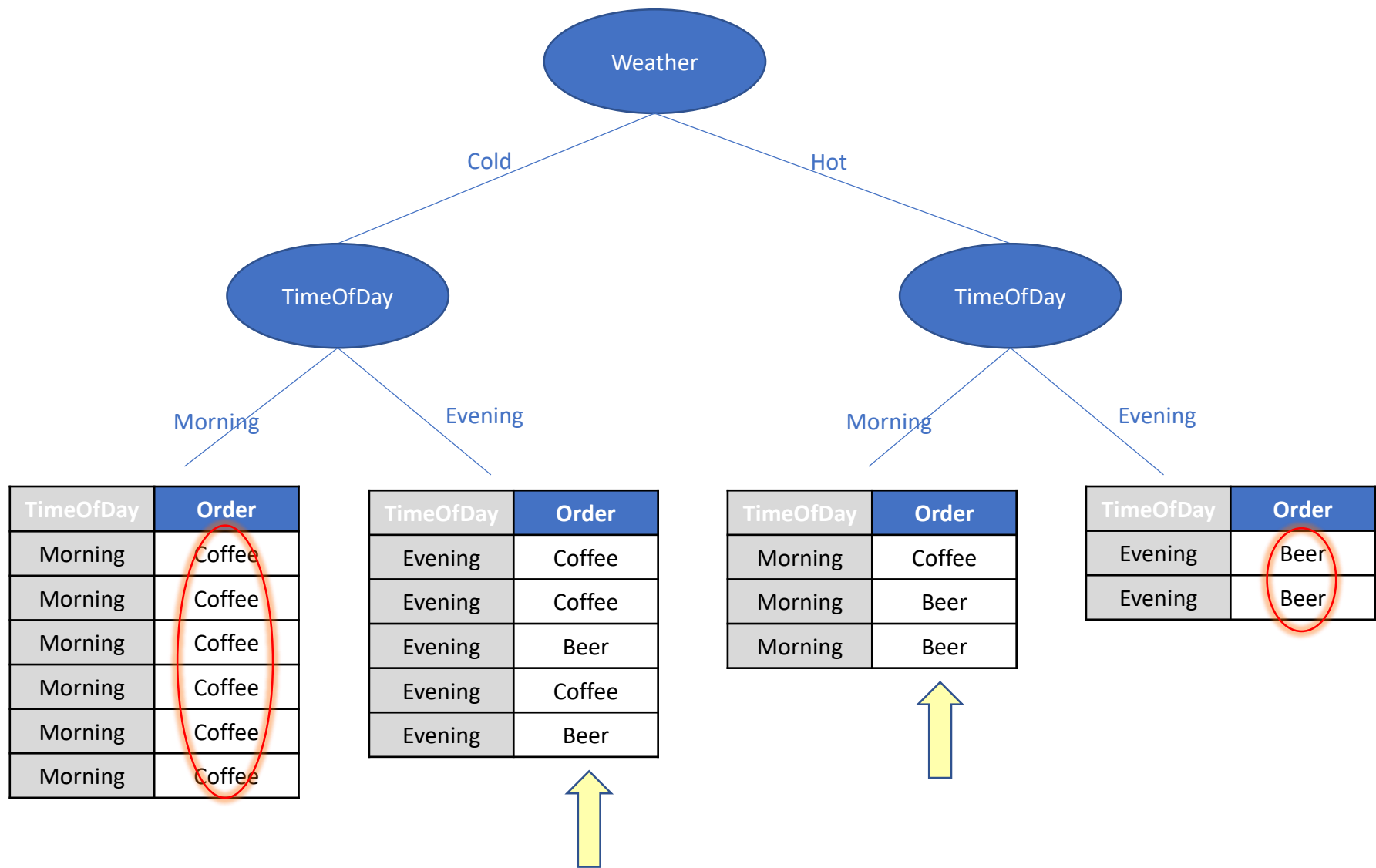
$\text{InfGain}(\text{Weather}) = 0.258$ ←



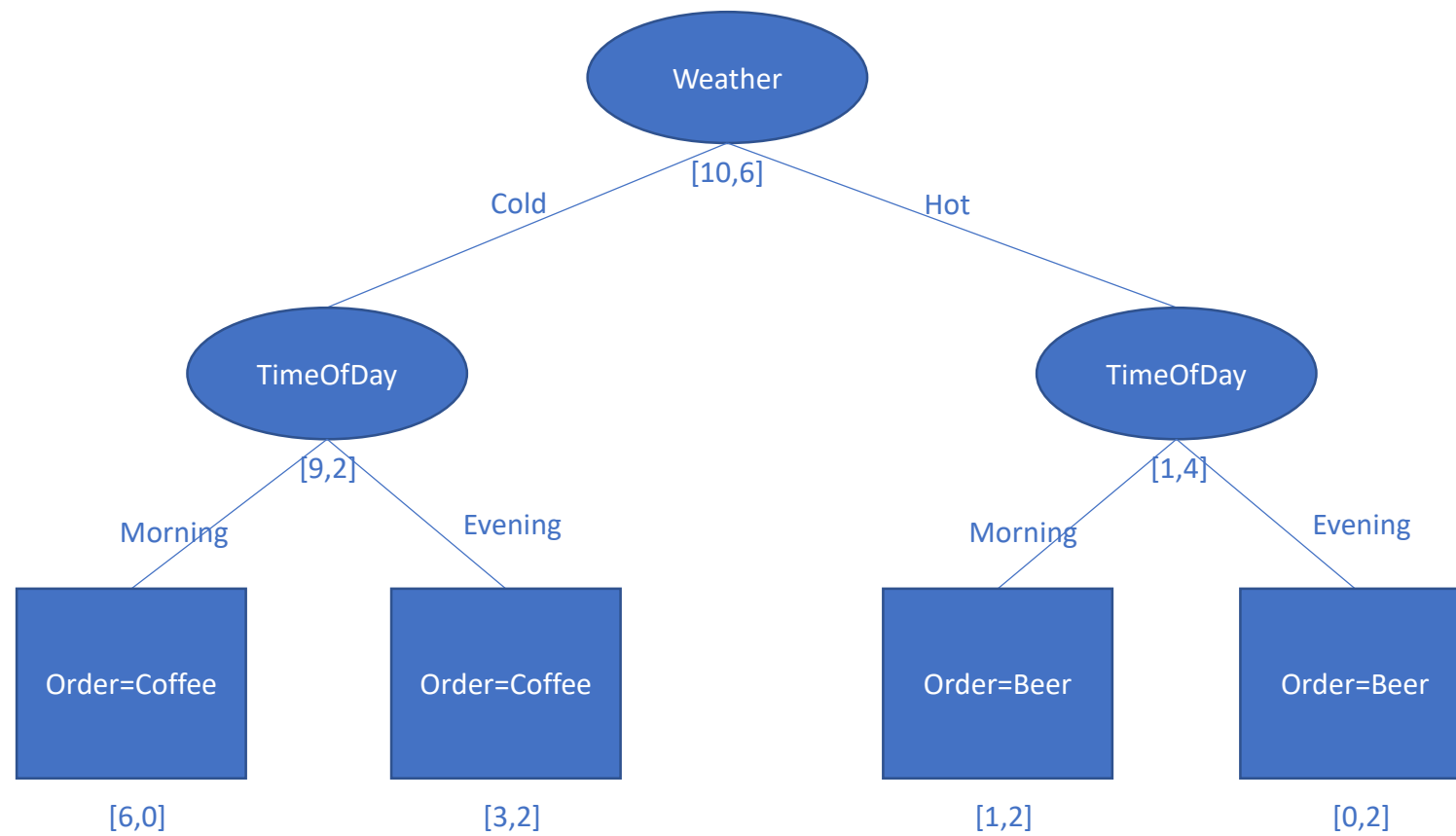
TimeOfDay	Weather	Order
Evening	Cold	Coffee
Morning	Cold	Coffee
Evening	Cold	Coffee
Evening	Cold	Beer
Evening	Cold	Coffee
Morning	Cold	Coffee
Morning	Cold	Coffee
Evening	Cold	Beer
Morning	Cold	Coffee
Morning	Cold	Coffee
Morning	Cold	Coffee

TimeOfDay	Weather	Order
Morning	Hot	Coffee
Evening	Hot	Beer
Morning	Hot	Beer
Evening	Hot	Beer
Morning	Hot	Beer

TimeOfDay je edini preostali atribut

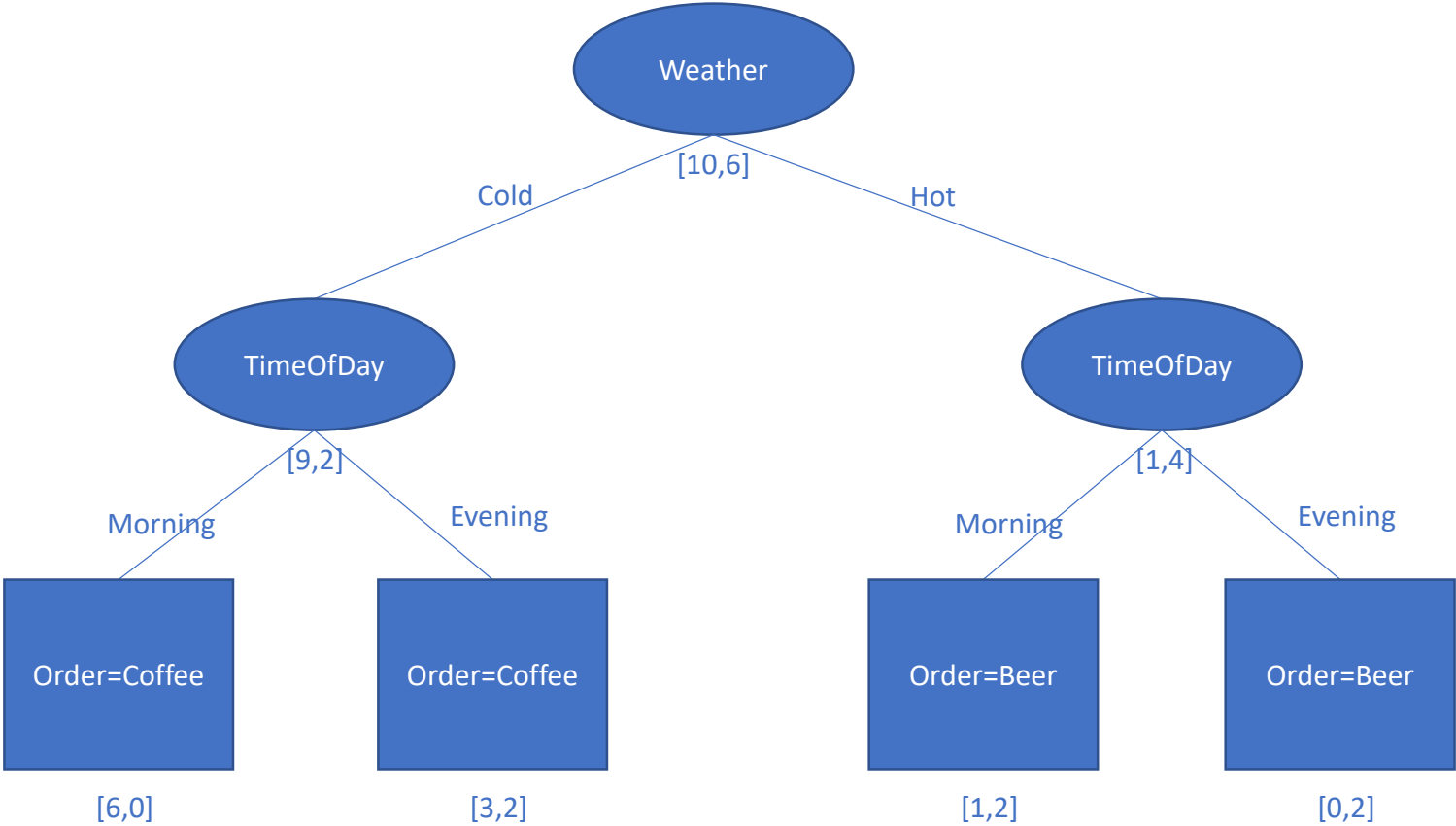


Ostali smo brez atributov za nadaljnjo delitev



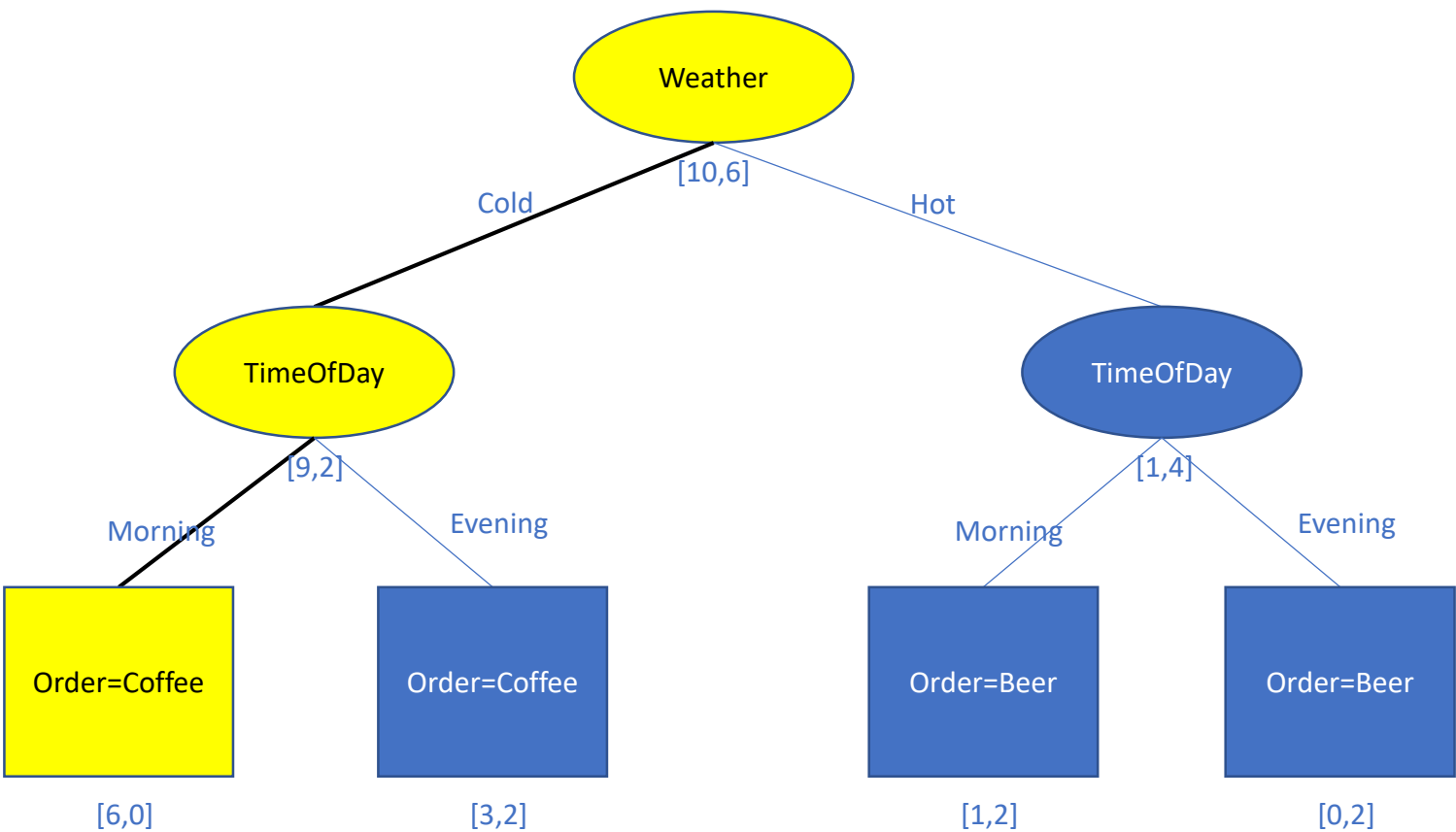
Naloga 1 (nadaljevanje)

Ovrednotite kvaliteto dobljenega drevesa s podano testno množico.



Testna množica:

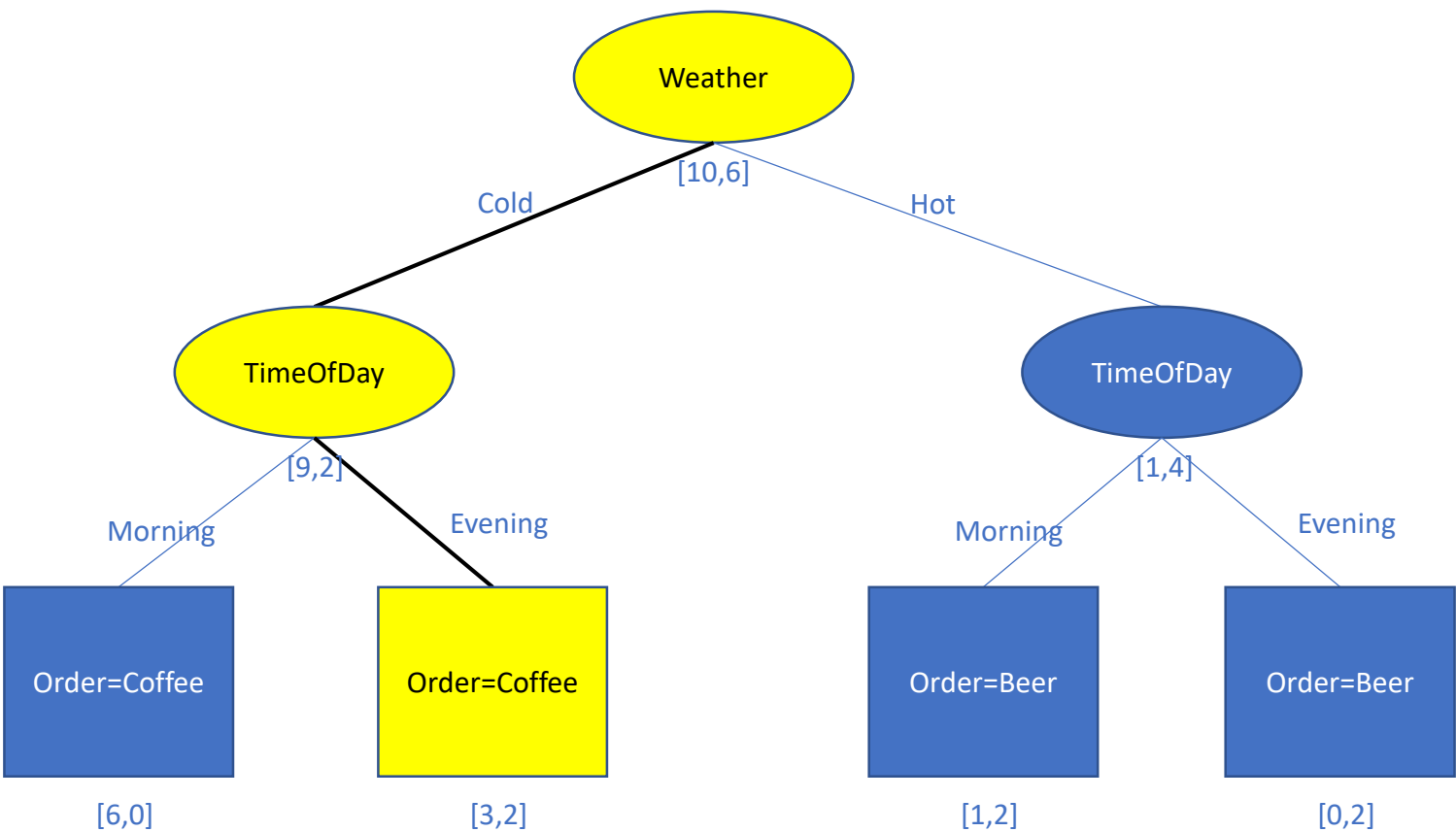
TimeOfDay	Weather	Order
Morning	Cold	Coffee
Evening	Cold	Beer
Morning	Hot	Beer
Morning	Cold	Coffee
Evening	Cold	Beer
Morning	Cold	Coffee
Evening	Hot	Beer
Evening	Hot	Beer
Morning	Cold	Coffee
Morning	Hot	Coffee
Morning	Cold	Beer
Evening	Cold	Coffee



Testna množica:

TimeOfDay	Weather	Order
Morning	Cold	Coffee
Evening	Cold	Beer
Morning	Hot	Beer
Morning	Cold	Coffee
Evening	Cold	Beer
Morning	Cold	Coffee
Evening	Hot	Beer
Evening	Hot	Beer
Morning	Cold	Coffee
Morning	Hot	Coffee
Morning	Cold	Beer
Evening	Cold	Coffee

Napoved: Coffee
Verjetnostna napoved: $6/6 = 1 = 100\%$ Coffee, $0/6 = 0 = 0\%$ Beer



Testna množica:

TimeOfDay	Weather	Order
Morning	Cold	Coffee
Evening	Cold	Beer
Morning	Hot	Beer
Morning	Cold	Coffee
Evening	Cold	Beer
Morning	Cold	Coffee
Evening	Hot	Beer
Evening	Hot	Beer
Morning	Cold	Coffee
Morning	Hot	Coffee
Morning	Cold	Beer
Evening	Cold	Coffee

Napoved: Coffee
Verjetnostna napoved: $3/5 = 0.6 = 60\%$ Coffee, $2/5 = 0.4 = 40\%$ Beer

		Napovedani	
		P	N
Dejanski	P	TP	FN
	N	FP	TN

Klas. točnost = $\frac{TP+TN}{TP+FP+FN+TN}$

Senzitivnost = $\frac{TP}{TP+FN}$

Specifičnost = $\frac{TN}{FP+TN}$

Preciznost = $\frac{TP}{TP+FP}$

F1 = $2 * \frac{Preciznost*Seznitivnost}{Preciznost+Senzitivnost}$

Ko obdelamo vse testne primere...

Pravi	Napovedani
Coffee	Coffee
Beer	Coffee
Beer	Beer
Coffee	Coffee
Beer	Coffee
Coffee	Coffee
Beer	Beer
Beer	Beer
Coffee	Coffee
Coffee	Beer
Beer	Coffee
Coffee	Coffee

		Napovedani	
		Coffee	Beer
Pravi	Coffee	5	1
	Beer	3	3

Klas. točnost = $\frac{8}{12} = 0.667$

Senzitivnost = $\frac{5}{6} = 0.833$

Specifičnost = $\frac{3}{6} = 0.5$

Preciznost = $\frac{5}{8} = 0.625$

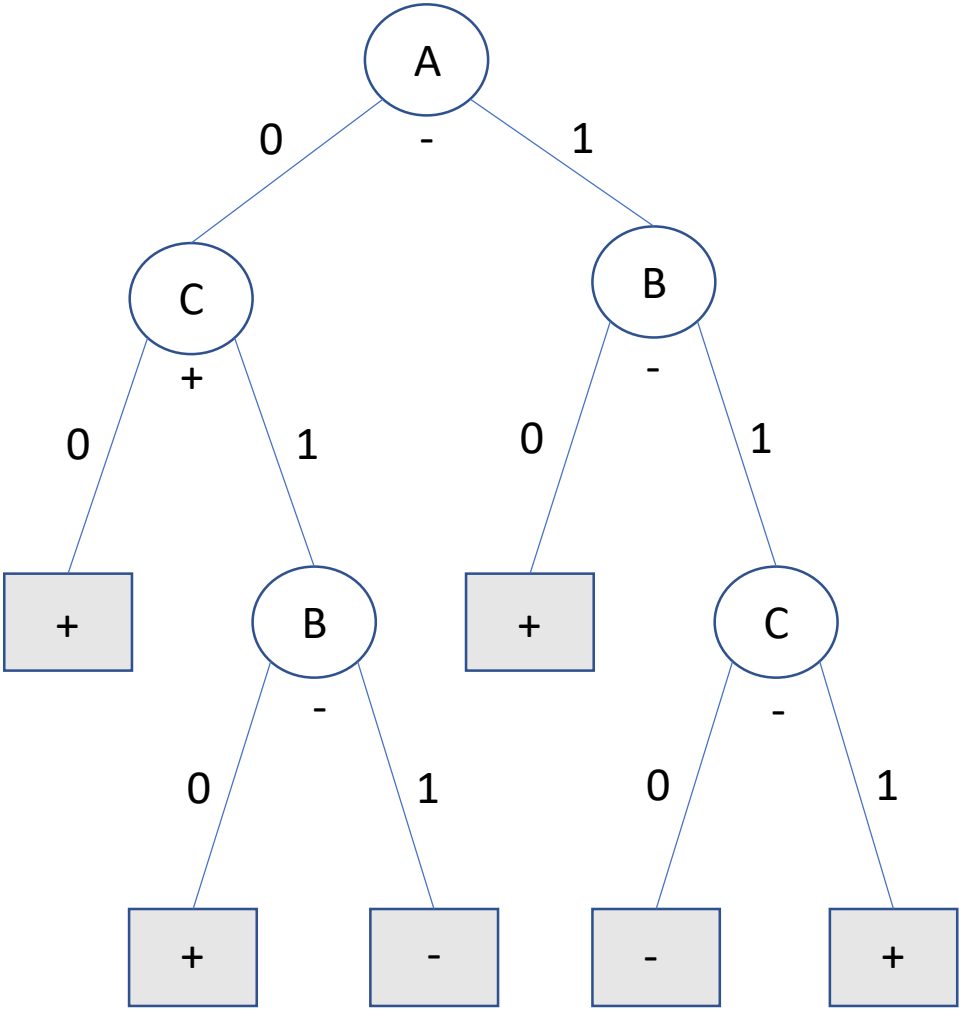
F1 = $\frac{2*0.625*0.833}{0.625+0.833} = 0.714$

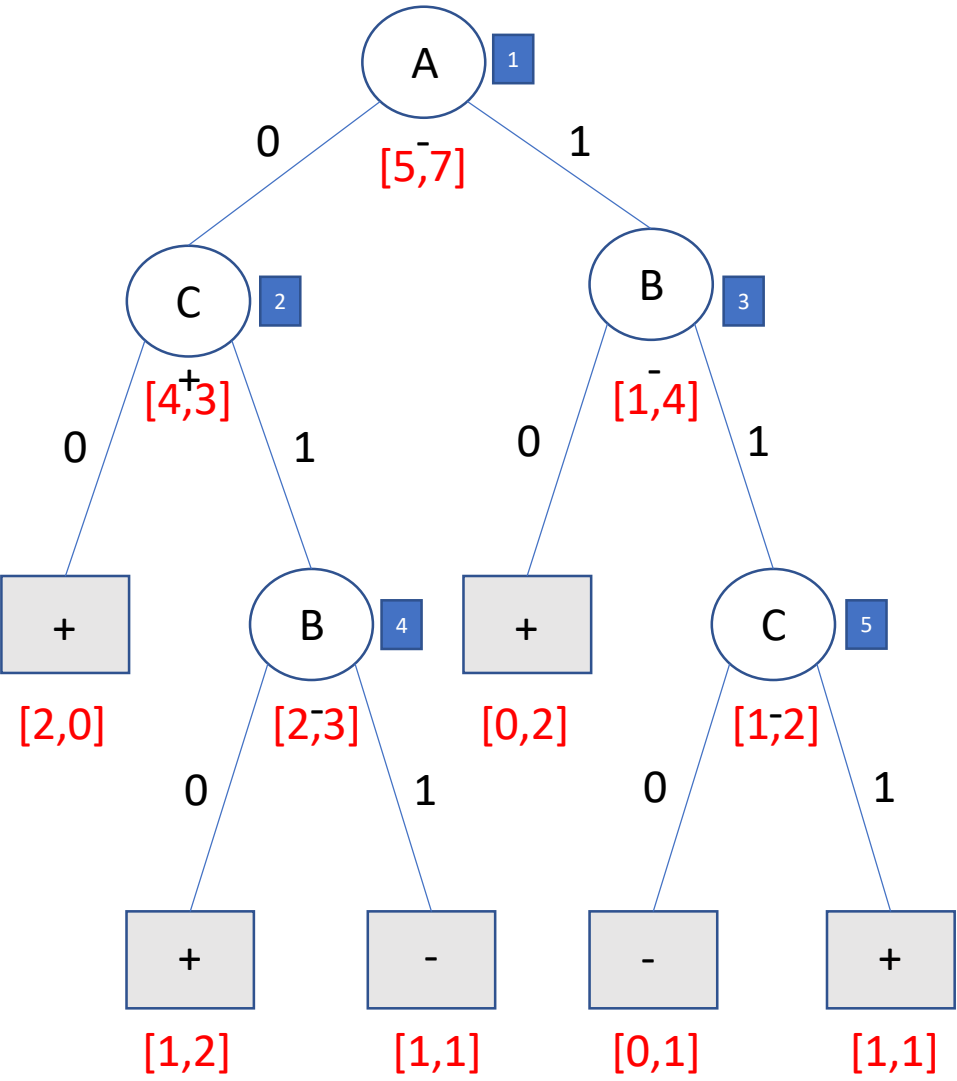
Naloga 2

Dano je odločitveno drevo in rezalna množica. Z rezanjem po metodi zmanjševanja napake (REP – Reduced Error Pruning) porežite dano drevo.

Rezalna množica:

A	B	C	R
0	0	0	+
0	1	0	+
0	0	1	+
0	0	1	-
0	0	1	-
0	1	1	+
0	1	1	-
1	0	0	-
1	0	1	-
1	1	0	-
1	1	1	+
1	1	1	-





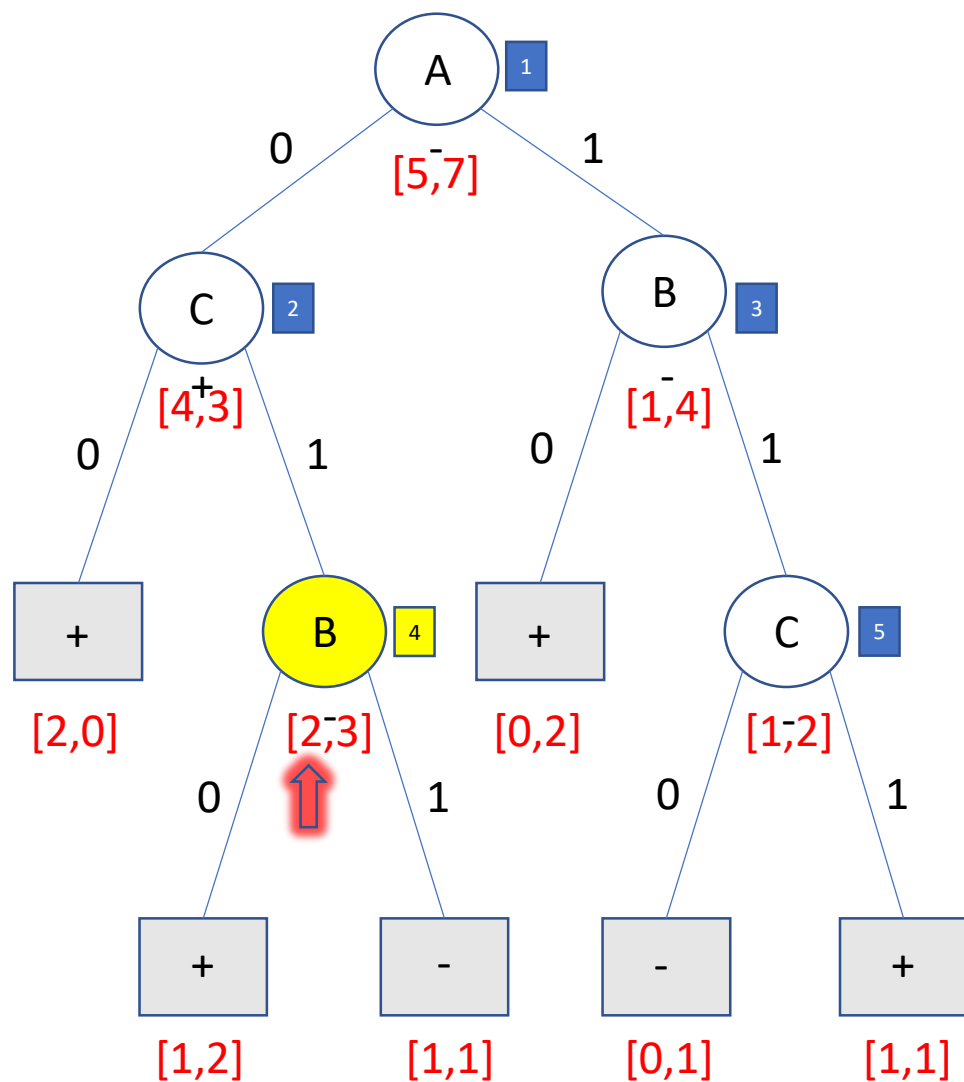
vsakemu vozlišču najprej pripišemo porazdelitev razreda iz rezalne množice

notranja vozlišča oštevilčimo 1-5 zaradi lažje identifikacije

$$e_v(4) = 2$$

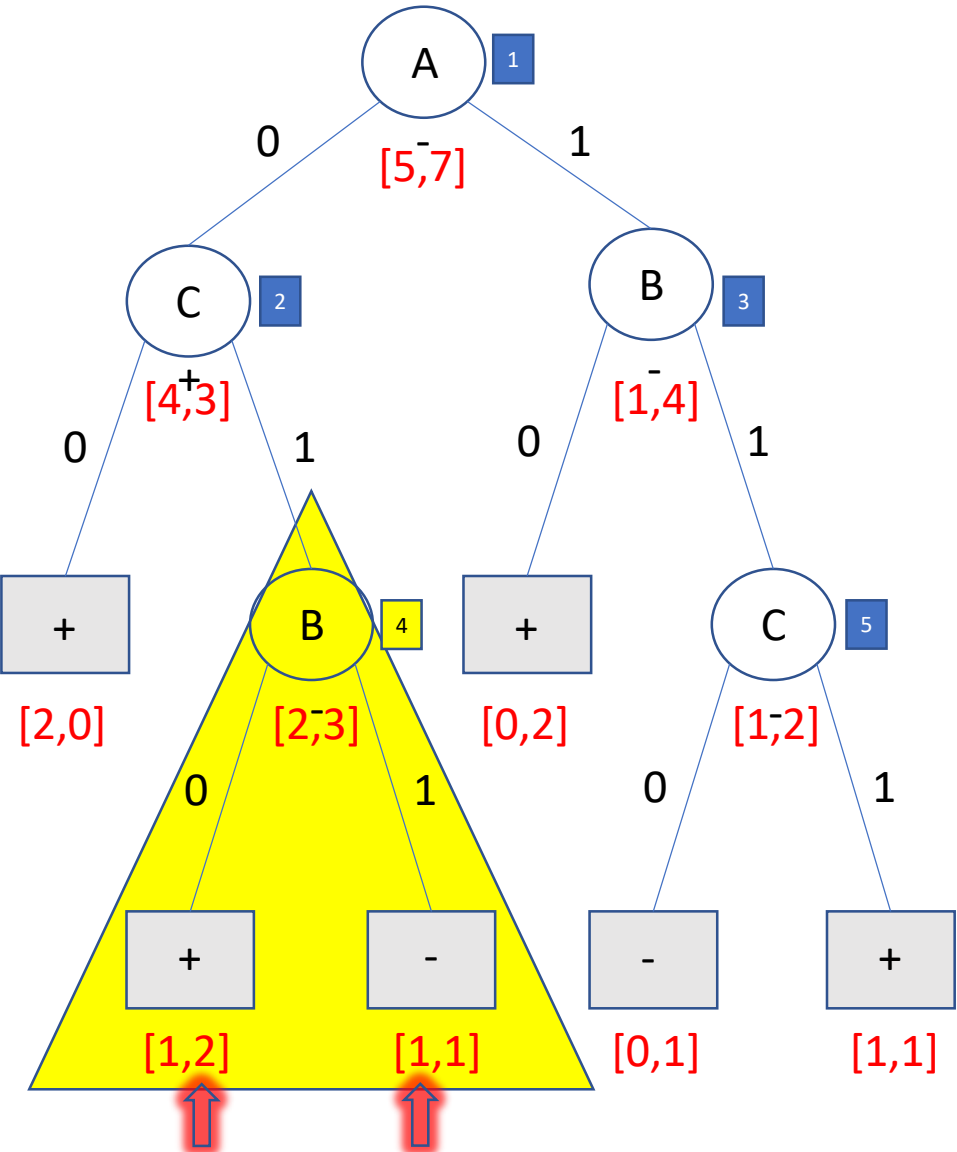


Število napak v opazovanem notranjem vozlišču, če bi drevo tam porezali

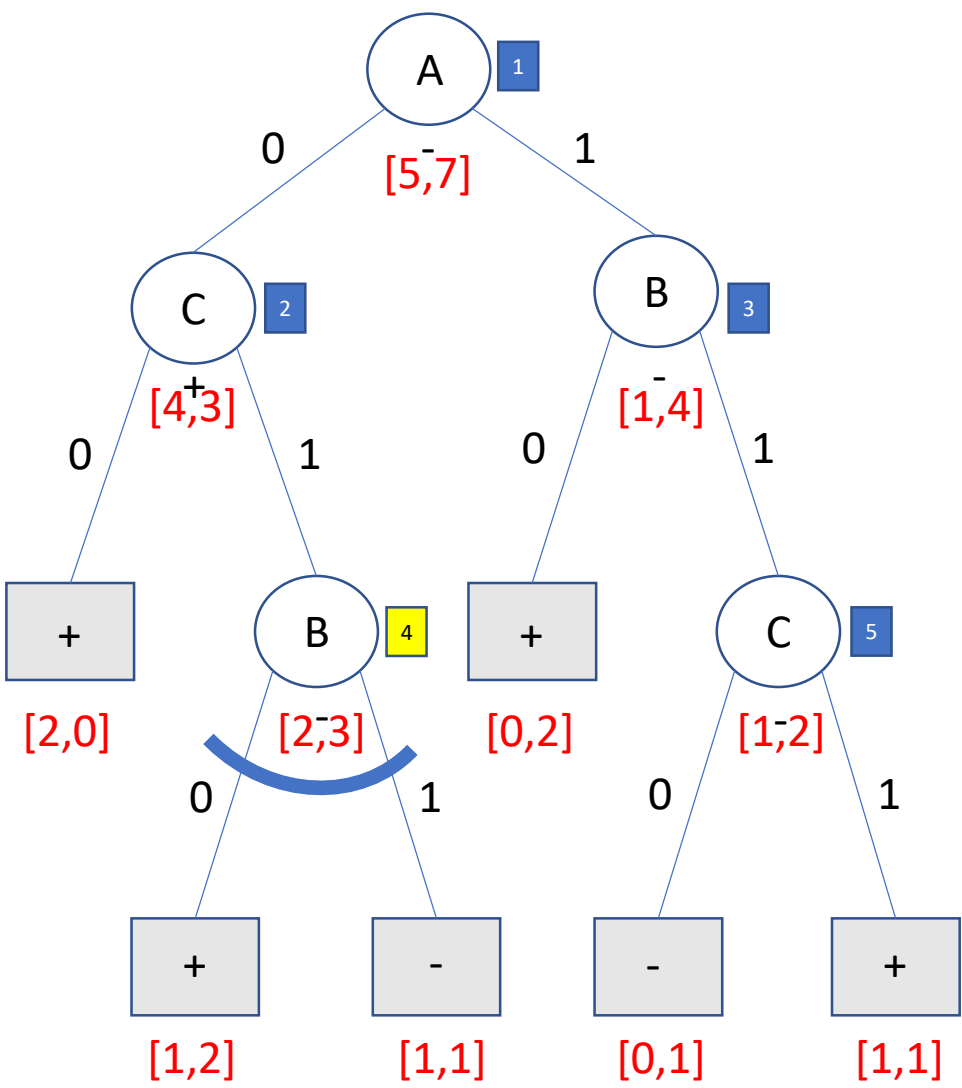


$e_v(4) = 2$
 $e_T(4) = 2+1=3$

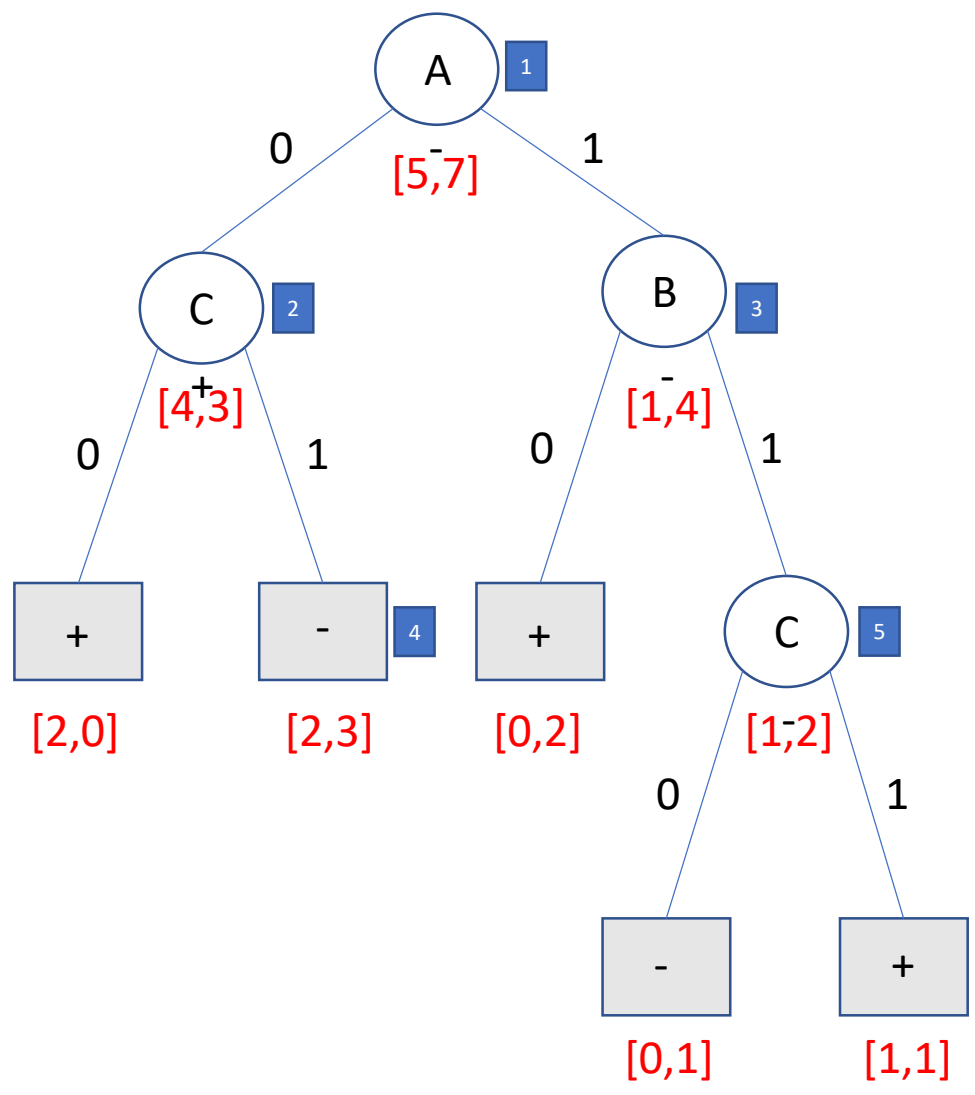
Število napak v
poddrevesu s korenom
v opazovanem
notranjem vozlišču



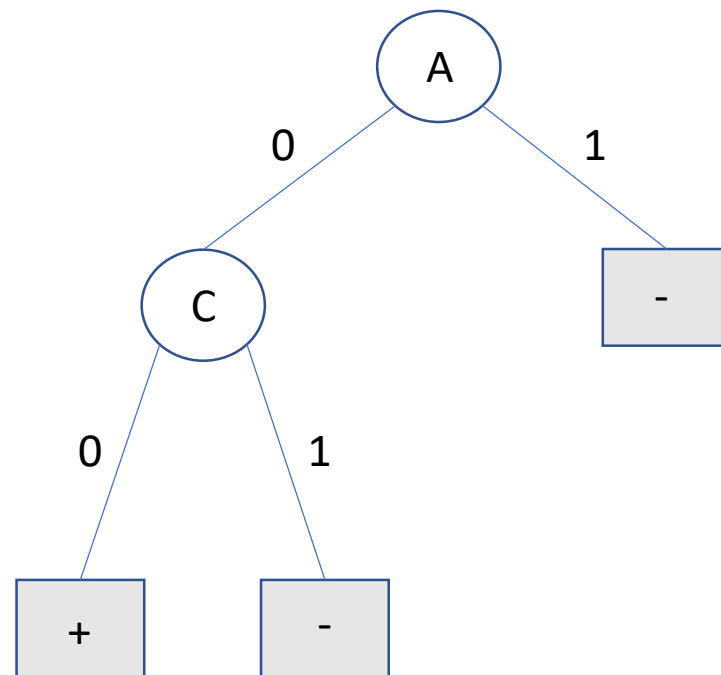
$e_v(4) = 2$
 $e_T(4) = 2+1=3$
 $G(4) = 3-2=1 \geq 0 \Rightarrow$ REŽEMO!



Dobitek rezanja.
Režemo takrat, ko je
 $G(v) \geq 0$



Porežite drevo do konca!



To je končno drevo.